Daniel Morales

djmorales2@miners.utep.edu | www.linkedin.com/in/daniel-morales-557644304 | danielsongit.github.io

Professional Summary

Mechanical Engineering student at UTEP with a 3.76 GPA and hands-on experience in CAD design, rapid prototyping, and hardware-software integration across robotics and aerospace-adjacent systems. Active contributor to a competitive fixed-wing aircraft design team with exposure to avionics and propulsion systems. Portfolio of engineering projects available at danielsongit.github.io.

Experience

Peer Educator Tutor Sep. 2026 – Present
University of Texas at El Paso
- Develop individualized study plans and tutor students in Introductory Mechanics, Calculus 1, and Calculus 2

Projects

Stress-Strain Plotter / Material Behavior Jan. 2026 – Feb. 2026
Individual Project
- Built a Python program to compute stress-strain relationships from raw data and generate plots using matplotlib
- Structured data processing using arrays and object-oriented design

Greenhouse Project Aug. 2023 – May 2024
Project Manager
- Led a 5-member team to design, build, and maintain a solar-powered irrigation system
- Coordinated task delegation and system implementation across the project lifecycle
- Improved plant growth efficiency by implementing automated watering schedules

Bell Advanced Vertical Robotics Aug. 2022 – Dec. 2022
Programmer and Assembly Assistant
- Independently programmed and calibrated a quadcopter drone in Python for flight, visual, and laser output based on computer input
- Modeled and fabricated drone components using CAD and 3D printing

Vex Robotics Aug. 2021 – Feb. 2024
Designer and Builder
- Modeled and assembled modular robot components in Fusion 360
- Iterated design improvements across build cycles to optimize competition performance

Activities & Leadership

El Paso Sun Birds Design Build Fly Team Sep. 2026 - Present
Member, Avionics and Propulsion Team
- Contributing to parts selection, circuit development, and technical documentation for avionics and propulsion systems on a competitive fixed-wing aircraft design team at UTEP.

American Institute of Aeronautics and Astronautics (AIAA) Sep. 2026 - Present
Member
- Participate in chapter meetings, engineering workshops, and technical events at UTEP as an active member of the university chapter.

American Society of Mechanical Engineers (ASME) Feb. 2026 - Present
Member
- Participate in chapter meetings and engineering workshops at UTEP.

Skills

Simulation and Analysis: Ansys Fluent, MATLAB
CAD and Design: Fusion 360, SolidWorks, 3D Printing, Rapid Prototyping, CAD Modeling
Engineering: Mechanical Systems Design, Hardware-Software Integration
Software and Programming: Python, Java

Education

University of Texas at El Paso El Paso, Texas
BS Mechanical Engineering. GPA: 3.76. *Ongoing*
Relevant Coursework: Mechanics of Materials, Fluid Mechanics, Thermodynamics, Intro Computer Science

El Paso Community College El Paso, Texas
AA Multidisciplinary Studies. *May 2024*